

SFS Scholarship Portal

Testing Summary

Prepared by Marcus | 28 July 2026

1. Testing Approach

Testing was done in three rounds as the portal was built out, rather than all at once at the end. Each round combined manual checking with automated scripts, so tests could be re-run any time a change is made (not just once before launch). Two methods were used throughout:

- **Local test server + live database:** the site was run on a local test server and pointed at the real (live) database, so tests behave exactly like a real visitor would, without needing to touch the live website.
- **Automated browser testing:** automated browser tools (Playwright) opened the site in Chrome, Firefox, and Safari engines to click through forms and flows the same way a real user would, plus Google Lighthouse for performance/accessibility/SEO scoring.

Reusable Python scripts were written for each type of test (see Section 5) so the same checks can be run again in seconds after any future change.

2. Round 1 - Full QA Pass (18 July 2026)

A full pass covering every category on a standard pre-launch QA checklist.

Functional Testing

- **Pages & flows:** All 27 pages load correctly; every interactive flow (apply, donate, login, review, etc.) works.
- **Form validation:** The apply form and donation form correctly reject empty or invalid entries with error messages.
- **Link checking:** 425 internal links, images, and scripts checked - zero broken.
- **Database operations (CRUD):** Create, read, update, and delete all verified directly against the database, with no leftover test data.

Cross-Browser & Cross-Device

- **Cross-browser:** 9 key pages tested in Chrome/Edge, Firefox, and Safari engines - zero errors in any of them.

- **Responsive layout:** Checked at phone (390px), tablet (768px), and desktop (1280px) widths. Found and fixed two issues: no mobile menu, and the donate page cards not stacking properly on phones.
- **Real device testing: not yet done** - Automated tools simulate screen sizes but not a real phone/tablet - a quick check on an actual device is still worth doing.

Performance (Google Lighthouse, homepage)

Category	Score
Performance	65 / 100
Accessibility	88 / 100
Best Practices	96 / 100
SEO	91 / 100

Security

- **Authentication & authorization:** Found and fixed a critical issue where any new signup could gain full admin access and read every applicant's data - confirmed via test signup (could pull all 17 student records), then re-verified as fixed (blocked from admin portal, sees no other students' data).
- **Data privacy:** Donors can only see their matched student; students see only their own record; anonymous visitors cannot read applicant data.
- **SQL injection:** Not applicable - the database layer used doesn't allow raw SQL from the browser.
- **Cross-site request forgery:** Low risk - login uses secure tokens, not cookies.
- **SSL / HTTPS:** Confirmed the live site loads securely (HTTPS) automatically via Netlify.
- **Cross-site scripting:** Minor - a few admin-only screens display names in a way that's a low-risk internal issue, flagged for future cleanup.

Usability, Accessibility & Compatibility

- **Navigation:** Clear layout and calls-to-action; mobile menu confirmed working after the fix.
- **Accessibility (88/100):** Two minor items flagged: some low colour-contrast text, and pages could use a clearer landmark for screen readers.
- **Payment gateway:** Not built yet - donation forms record intent only, no real money changes hands yet.
- **Analytics:** Not connected yet - needs a Google Analytics account.
- **Database integrity:** Row-level security rules checked and confirmed correct for every user role.

SEO

- **Page basics:** Titles and mobile viewport settings confirmed correct on all pages.
- **Meta descriptions:** Were missing on all 10 website pages - added.
- **robots.txt & sitemap.xml:** Added so search engines can properly index the site.

3. Round 2 - Public Forms & Security Hardening (27 July 2026)

- **Public apply & donate forms:** Discovered the public application form and donation form could not submit at all, due to an unusual database-level restriction. Rebuilt both to go through a secure server-side function instead of a direct database write, then retested end-to-end with an automated browser - both forms now submit successfully and were confirmed working before this fix went live.
- **Document viewer:** Added a 'View' popup for uploaded documents in two places (the admin Documents page and the student profile panel) and tested that it correctly opens both images and PDF files inline.
- **Independent security review:** An external, informal security review (conducted by a developer known to Marcus) found that any newly signed-up account - with no verification required - could read the entire donor list and staff directory, and could add itself as committee staff to gain full admin access. This was fixed the same day it was found, and the fix was confirmed by directly checking the database's access-control rules: the self-promotion path is closed, and staff/donor data is now restricted to authorised staff only.

4. Round 3 - Account Security & Email Verification (28 July 2026)

- **Student account creation:** The student account creation process previously emailed a plaintext password directly, which caused Gmail to flag the message as suspicious/dangerous. Rebuilt so a staff member issues a secure one-time invite link instead - tested by generating an invite and confirming it correctly lands the student on a 'set your own password' page.
- **Donor email verification:** Rebuilt so a donor's account only becomes active after they click a confirmation link sent to their email. Verified end-to-end on the live site: register, see a 'check your email' screen, receive the email, click the link, and land on a working dashboard.
- **Confirmation email branding:** Customised the confirmation email to match the site's branding (colours, wording) instead of the plain default - tested by triggering a real signup and checking the email visually.
- **Document upload flow:** Ran a dedicated test that mimics a real applicant uploading a document (creates a test entry, uploads a sample file, saves the record) and confirmed it shows up correctly and opens properly in the admin portal's document viewer.

5. Reusable Test Scripts

These scripts live in the project folder and can be re-run any time after a future change, by serving the site locally and running them one by one:

Script	What it checks
<code>check_site.py</code>	All pages load; no broken links
<code>check_backend.py</code>	Backend rules & data privacy
<code>check_crud.py</code>	Create / read / update / delete against the database
<code>cross_device.py</code>	Cross-browser + responsive screenshots
<code>check_forms.py</code>	Form validation (apply & donate forms)
<code>check_mobile.py</code>	Mobile menu & layout
<code>browser_test.py</code>	Full login flows & security checks
<code>upload_test.py</code>	Document upload flow (mimics a real applicant)

6. Known Gaps - Not Yet Covered by Testing

- **Real device testing:** Automated tools simulate screen sizes; a quick check on an actual phone/tablet hasn't been done.
- **Load / stress testing:** Not run, to avoid overloading the free-tier database with artificial traffic.
- **Payment flow:** Cannot be tested until a payment gateway is chosen and set up.
- **Analytics tracking:** Not tested since it isn't connected yet.